Pediatric Asystole / PEA

PC-01 | Pediatric Cardiac

Quick flow

01 Pulseless Arrest [EMT / AEMT / Paramedic]

Confirm apnea/pulselessness, DNR/MOST, and obvious-death criteria; apply monitor/AED and check leads before treating apparent asystole.

02 High-Quality CPR [EMT / AEMT / Paramedic]

100-120/min • depth at least ⅓ chest AP diameter • full recoil • pauses under 10 sec • change compressor every 2 min.

03 Oxygenate + Ventilate [EMT / AEMT / Paramedic]

BVM first; 15:2 with two rescuers or 30:2 with one rescuer when no advanced airway; avoid hyperventilation.

04 IV/IO + Early Epinephrine [AEMT / Paramedic]

Obtain rapid vascular access without delaying CPR; give 0.01 mg/kg IV/IO every 3-5 min.

05 Reversible Cause? [EMT / AEMT / Paramedic]

Hypoxia • hypovolemia • acidosis • temperature • metabolic/electrolyte cause • tension • toxins • thrombosis.

06 ROSC? [EMT / AEMT / Paramedic]

Yes: PC-08 Pediatric Post-Resuscitation. No: continue 2-min cycles, reassessment, and treatment of cause.

Overview

- For infants and children after the newly born period, asystole and PEA are commonly the end result of hypoxia or respiratory failure; ventilation and high-quality CPR are central interventions.
- Search aggressively for reversible causes while minimizing interruptions in compressions.

Provider-Level Actions

Use a length- or weight-based pediatric resuscitation system and perform immediate lifesaving care without delaying oxygenation, ventilation, CPR, or defibrillation.

EMT

- Begin continuous high-quality CPR, apply AED/monitor, provide oxygenation/ventilation, and use 15:2 with two rescuers or 30:2 with one rescuer when no advanced airway is present.
- Use a pediatric length-based resuscitation system, change compressors every 2 minutes, and limit rhythm/pulse checks to less than 10 seconds.

AEMT

- Perform all EMT care plus IV/IO access, weight-based epinephrine, glucose analysis, crystalloid when hypovolemia is suspected, and authorized advanced airway support.
- Do not delay CPR or ventilation for repeated IV attempts; use IO promptly when IV access is not rapidly available.

Paramedic

- Perform all prior care plus rhythm confirmation, continuous waveform $EtCO_2$, advanced reversible-cause treatment, and post-ROSC transition.
- Do not interrupt compressions to intubate; use effective BVM or an authorized BIAD when it reduces pauses.

Indications

- Pediatric patient in pulseless arrest with asystole or pulseless electrical activity.

Treatment steps

- Asystole/PEA is nonshockable. Continue CPR and treat reversible causes; defibrillate only if the rhythm changes to VF/pulseless VT.
- With an advanced airway, use continuous compressions and ventilate 20-30/min (1 breath every 2-3 seconds). Avoid hyperventilation.
- Check glucose, temperature, and likely reversible causes during CPR: hypoxia, hypovolemia, acidosis, hypo/hyperkalemia or metabolic disturbance, hypothermia, tension pneumothorax, tamponade, toxins, and thrombosis.
- Use continuous waveform $EtCO_2$ when available. Use trends to improve CPR; an unexpected sustained increase may indicate ROSC, but do not use a single $EtCO_2$ value alone to make termination decisions.
- Perform rhythm/pulse checks no more than every 2 minutes and keep each pause under 10 seconds.
- After ROSC, move immediately to PC-08 Pediatric Post-Resuscitation Care; maintain airway support, oxygenation, ventilation, glucose/temperature management, blood pressure support, and early notification.

Medications

Epinephrine 0.1 mg/mL (1:10,000)

0.01 mg/kg IV/IO as early as possible; maximum single dose 1 mg; repeat every 3-5 minutes

Normal saline

10-20 mL/kg IV/IO only when hypovolemia is suspected; reassess perfusion, lungs, and suspected cause after each bolus

Warnings / precautions

- Do not hyperventilate or pause compressions for endotracheal intubation.
- IV/IO access and medications are secondary to CPR and ventilation.
- Do not routinely give calcium or sodium bicarbonate unless a specific indication exists.
- Do not delay transport for repeated airway attempts.

Clinical pearls

- Pediatric nonshockable arrest is often hypoxic/asphyxial; effective ventilation is a central intervention.
- Use waveform $EtCO_2$ trends to improve CPR quality and detect possible ROSC, not as a stand-alone termination decision.

Related pathways / contacts

AR-05 Pediatric Airway - Open for pediatric BVM and airway support.

AR-06 Pediatric Failed Airway - Open for ineffective oxygenation/ventilation or unsafe further airway attempts.

PC-02 Pediatric Bradycardia / Poor Perfusion - Open if a pulse returns but bradycardia with poor perfusion persists.

PC-08 Pediatric Post-Resuscitation Care - Open immediately after ROSC.

AO-02 Newly Born - Open for neonatal resuscitation immediately after birth.

Source and references

Claiborne County EMS approved protocol manual. | Current Tennessee EMS scope of practice and Claiborne County standing orders control for all provider permissions.

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